

Non-invasive and Adaptive Brain-Computer Interface Based on Electroencephalographic Signals from the Primary Motor Cortex

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Abstract— Brain-Computer Interfaces (BCIs) represent one of the most innovative frontiers in the field of applied Neurology. The engineering interest surrounding this type of device is mainly focused on the human-device connection and the analysis of electroencephalographic (EEG) signals, finding applications in the treatment of several neuropathologies that may impair an individual's voluntary movement. Among the main clinical conditions addressed by the applicability of the present project are Locked-in Syndrome (LS), Amyotrophic Lateral Sclerosis (ALS), multiple sclerosis (MS) and other types of neurotrauma capable of causing irreversible damage to the spinal cord. The development of an integrated environment, combining hardware and software components, therefore enables the creation of new communication modalities, placing versatility, practicality and patient well-being at the core of system use.

Keywords— *Primary Motor Cortex, Brain-Computer Interface, Electroencephalography, Signal Processing, Machine Learning*

I. INTRODUCTION

In the neurological domain, pathologies and traumatic injuries affecting the Central Nervous System (CNS) or the spinal cord can impair motor signal transmission and disrupt communication between the brain and voluntary muscles. However, in many cases brain activity remains partially or fully intact, allowing the extraction of meaningful information. BCIs represent a promising solution for establishing alternative communication pathways such as the control of external devices, with particular relevance to signals acquired from the Primary Motor Cortex (M1), which is involved in motor command generation and directly connected to lower motor neurons and voluntary muscles [1]. In this context, the present project aims to develop a non-invasive and adaptive EEG-based BCI to improve the accessibility and practicality of such systems in real-world scenarios where M1 functionality is preserved, leveraging the safety and versatility of EEG technology. To address the associated challenges, the proposed system integrates EEG signal processing techniques with dedicated software environments for acquisition, analysis, and training, enabling progressive adaptation to the patient. The system is designed with particular attention to usability and potential integration into clinical and rehabilitation settings, representing a foundational solution toward more complex human-machine applications.

II. SYSTEM OVERVIEW

A. Hardware and Signal Acquisition

The components are designed to acquire the electroencephalographic signal from the M1 area, while minimizing electronic noise, motion artifacts and noise related to the electrode-scalp interface, thereby optimizing the Signal-to-Noise Ratio (SNR). Signal acquisition relies on soft dry conductive-polymer electrodes with high conductivity, suitable for experimental EEG applications. These electrodes are made of thermoplastic elastomer with Ag/AgCl terminals, ensuring high reproducibility and reliability in biopotential measurements [2]. This choice is motivated by their ease of use in scenarios involving frequent and long-term application, improving patient comfort without compromising signal quality, which is comparable to gel-based electrodes one. The reference electrode is an earclip electrode, also with Ag/AgCl terminals, placed on the ipsilateral earlobe [3]. The electrode placement follows the International 10–20 System, restricted to the M1 area and the cortical region of the dominant hand. It includes one electrode at Cz over the central sulcus, one at FC3/FC4 in relation to motor imagery and one at C3/C4 to focus on the hand motor cortex area. The electrodes are mounted on a support band to avoid the use of a full cap and to improve user comfort. The connection to the board is established via 0.5 m snap electrode cables, and the device used is a 4-channel OpenBCI Ganglion Board, chosen for its ease of use and its compatibility with a fixed device for signal processing [4]. The board can be placed either posteriorly at the level of the C7 vertebra or on the ipsilateral side in the medial region of the biceps brachii, using an elastic strap. Communication with the device is established via a Bluetooth Low Energy (BLE) connection enabled by a dedicated USB dongle.

B. Software and Processing Architecture

The board communicates with the device by acquiring EEG signals from the scalp and transmitting them to the software for processing. This part of the system is divided into two related procedures. The first concerns the study of the EEG signal and the training of the user. The software used is the OpenBCI GUI [5], which enables real-time visualization of the signals from individual electrodes, allowing verification of correct placement and identification of possible artifacts. The signal is then analyzed in MATLAB using the EEGLAB and ERPLAB toolboxes, which enable data visualization and analysis. The second part of the software is dedicated to device control, based on the acquired signals. BrainFlow [6] is the principal platform, integrated after the verification of signal quality. It is developed in Python, enabling the development of dedicated software for controlling the device cursor.

III. METHODOLOGY

A. EEG Signal Analysis and Training Procedure

The system operation begins with an initial phase devoted to the identification of the EEG event on which command recognition is based. More specifically, the final system is based on Event-Related Potentials (ERPs), a type of evoked potentials (EPs). Indeed, ERPs mainly appear as brain responses to auditory, visual, or tactile stimuli [7]. In our case, the focus is on the P300 signal, a wave component associated with specific cognitive functions, which in turn consists of an overlap of two subcomponents, P3a and P3b. P300 occurrence requires a neural event, and placing the electrodes along the C line allows the ideal extraction of these signals. Therefore, the research is based on motor imagery tasks, during which the patient is asked to imagine moving their dominant hand. More specifically, the movement is divided into four categories: up, down, right, and left. The variation in the occurrence of P300 signals in the patient will be analyzed as a function of the imagined hand movement direction. This process is made possible through real-time EEG signal monitoring using the dedicated OpenBCI GUI. For each imagined direction, the data are recorded and later analyzed in a post-acquisition phase using MATLAB, which support the identification of ERPs. Each patient will exhibit a specific pattern of these signals. Therefore, their data will be stored for characterization. Finally, another part of the signal analysis is also focused on selecting objects displayed on the device screen. In fact, the selection is achieved by stopping the motor imagery process and closing the eyes for a specific period of time. This latter event, is distinguishable in the EEG signal, and, as with cursor directions, it will be analyzed in the post-acquisition phase in order to identify the corresponding pattern from which it originates.

B. Interface Control and User Interaction

The commands and functionalities available to the user, are mainly related to controlling the cursor on the screen of the connected device. The range of motion covers 360 degrees and, is composed by four basic movements mentioned before. The system is accessed through the software launch, which can be performed by the user or by the operator assisting the patient. Once the software is running, the user must rely on their motor imagery, developed and consolidated during the training phase. The available functionalities are the same as those accessible to any user when interacting with their device. Therefore, the usage session continues as long as the user wishes. Patients can move the cursor, select objects, and type using a dedicated on-screen keyboard, if required. Moreover, they can navigate both local windows and online web content. In conclusion, at the end of the session, the patient can terminate it by pressing a dedicated command, restoring control to traditional input devices.

IV. FUTURE DEVELOPMENTS

The true potential of the present project primarily lies in its future developments and subsequent evolutions. With regard to hardware improvements, the main innovations will primarily concern the elements in direct contact with the patient. Traditional electrodes will in fact be replaced by innovative biopolymer electrodes featuring a single patch-like structure, capable of improving both the quality of the acquired signal and the user's comfort. The patch will be designed to adapt to the patient's anthropometric characteristics and to precisely cover the interested scalp

regions. It will feature an "L-shaped" configuration in order to ensure uniform coverage of the Cz, C3/C4, and FC3/FC4 positions. Then, communication between the electrodes and the processing board will remain wired. However, thinner and more flexible cables will be used, allowing better adaptation to the patient's movements and neck morphology, while the board itself will continue to be preferably positioned near the C7 vertebra. Communication between the board and the user's device will instead remain wireless, based on the same technology previously introduced, but without the need for a dedicated USB dongle. In parallel, the software environment will evolve towards a more versatile and accessible configuration, structured into a standalone application, with two functions: will be intended for clinicians and professionals to support the initial phases of EEG signal analysis and interpretation, while the second will be designed for the end user and will directly communicate with the BCI system. The latter will be available across multiple platforms, including computers and mobile devices. On smartphones and tablets, this application will implement a virtual cursor controlled via neural signals, enabling command selection and interaction with the interface. Furthermore, given the complexity of operations such as web browsing, an artificial intelligence agent will be integrated to actively assist the user, simplifying interaction and improving the overall user experience. The system will be designed to ensure high interoperability, allowing any device equipped with the application to connect to the BCI system, without restricting the patient to the use of a single terminal. At the same time, the application will include dedicated functionalities for clinicians to remotely monitor patient parameters and the quality of the acquired signal, with the possibility of transmitting relevant data to a control platform for the evaluation of progress and the implementation of any corrective actions. In conclusion, the objective is the development of an integrated ecosystem that enables patients to communicate more effectively, flexibly and autonomously with devices and other individuals, facilitating activities such as message sending, access to web information and the maintenance of social relationships. This hardware and software ecosystem will be named "SyNexus", inspired both by the synaptic connections within the nervous system and by the Latin term "Nexus", symbolizing the bond and interconnection that the system aims to establish.

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